

EQ Prerequisite: Minimum Acoustic Performance

This prerequisite applies to

- ▶ BD+C: Schools

INTENT

To provide classrooms that facilitate teacher-to-student and student-to-student communication through effective acoustic design.

REQUIREMENTS

SCHOOLS

HVAC Background Noise

Achieve a maximum background noise level of 40 dBA from heating, ventilating, and air-conditioning (HVAC) systems in classrooms and other core learning spaces. Follow the recommended methodologies and best practices for mechanical system noise control in ANSI Standard S12.60-2010, Part 1, Acoustical Performance Criteria, Design Requirements and Guidelines for Schools or a local equivalent for projects outside the U.S.

Exterior Noise

For high-noise sites (peak-hour Leq above 60 dBA during school hours), implement acoustic treatment and other measures to minimize noise intrusion from exterior sources and control sound transmission between classrooms and other core learning spaces. Projects at least one-half mile (800 meters) from any significant noise source (e.g., aircraft overflights, highways, trains, industry) are exempt.

Reverberation Time

Design each of the classrooms and other core learning spaces to meet one of the following:

- ▶ The total surface area of acoustic wall panels, ceiling finishes, and other sound-absorbent finishes equals or exceeds the total ceiling area of the room (excluding lights, diffusers, and grilles). Materials must have an NRC of 0.70 or higher to be included in the calculation.
- ▶ The calculated or measured reverberation times comply with ANSI Standard S12.60-2010, Part 1.
- ▶ For spaces $\geq 20,000$ Cubic Feet (566 Cubic Meters), the calculated or measured reverberation times comply with Table 1:

Table 1. Maximum reverberation time for large volume spaces

Type of room	Mid-frequency Reverberation time (s)
Large lecture room (more than 50 people)	≤ 1.0
Library	≤ 1.0
Assembly hall, multi-purpose hall	0.8 ~ 1.2
Indoor sports hall, swimming pool	1.5~2.0
Gymnasium/activity studio	≤ 1.5
Dance studio	≤ 1.2

*For details of the above reverberation time requirements, please refer to Building Bulletin 93 – Acoustic Design of Schools: Performance Standards published by the UK's Department for Education in February 2015.

Note: the reverberation time is quoted in terms of the mid-frequency reverberation time, T_{mf} , which is the arithmetic average of the reverberation times in 500Hz, 1 kHz, and 2kHz octave bands, or the arithmetic average of the reverberation times in the one-third octave bands from 400 Hz to 2.5 kHz.

Exceptions

Exceptions to the requirements because of a limited scope of work or to observe historic preservation requirements will be considered.

For spaces which involve complex sound calculations and/or special considerations, such as those for music rehearsal/performance, recording studio, drama studio, teaching space intended specifically for students with special hearing or communication needs, this Prerequisite shall not apply and engagement with an acoustician is critical to the acoustic design and performance of the space.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The reverberation time requirements for large-volume spaces have been updated to replace the standard: NRC-CNRC Construction Technology Update No. 51 with a table developed by USGBC based on UK Department for Education performance standard Building Bulletin 93.

The ASHRAE and AHRI references for HVAC background noise have been removed to emphasize ANSI Standard S12.60-2010, Part 1, as the primary source for background noise in classrooms and other core learning spaces. Background noise levels can still be calculated or measured using the ASHRAE and AHRI references.

Exceptions for certain spaces are included to clarify the scope (and limitations) of this prerequisite.

Step-by-Step Guidance

Steps 1-9. Refer to the LEED v4 reference guide. Note that for step 4, it is still acceptable to verify background noise via calculations from the ASHRAE Handbook or AHRI standard, or measurements via ANSI Part 1 annex A1 or ASHRAE Handbook, or other equivalent method.

Step 9. Select reverberation time compliance option for each classroom and core learning space

In addition to the language in this step in the LEED v4 reference guide, add:

Option 3. Maximum reverberation time for large volume spaces

Confirm that the reverberation time for the space complies with Table 1 in the rating system using calculations or measurements. Retain calculations or measurement logs for credit documentation.

Step 10. Remove this step (covered above in step 9, option 3).

Further Explanation

Connection to Ongoing Performance

- ▶ LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to minimize HVAC background noise, exterior noise and reverberation time for newly constructed schools can help contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.